

On the Methodological Foundations of Modern Microeconomics: Frank Knight and the “Cost Controversy” in the 1920s

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The criticism of Marshallian economics in the 1920s occupies a central position in the development of modern microeconomic theory. Crucial to that criticism, historians widely believe, was the so-called cost controversy—the critical debate that took place between 1926 and 1930 principally on the pages of the *Economic Journal*, in articles by Sraffa, Pigou, Robertson, Shove, Robbins, Young, and Schumpeter. The cost controversy is seen as a caesura in the discussion of Alfred Marshall's economics, and it is viewed as a prelude to the theory of imperfect competition.

This interpretation, however, unnecessarily narrows what is, in fact, a rich and broad debate. It overlooks a major theoretical event of those years, an event that the cost controversy was only a part of: the foundation of the concept of pure competition of modern microeconomics, which led to the separation between statics and dynamics, which in turn enabled economists to overcome the analytical difficulties of Marshall's microeconomics. It is within this broad perspective that we may

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appreciate the absolute centrality of one contribution that the historical reconstruction has generally treated as marginal and has partially neglected: the work of Frank Knight on price theory in the period 1921–25.

This article reconstructs Knight's contribution to price theory in the 1920s. It places Knight's work in the context of the ongoing English side of the debate on Marshall's theory. In the first section we discuss the key theoretical problem of Marshall and the Marshallians up until A. C. Pigou's *Wealth and Welfare* (1912): the dilemma of "increasing returns and competition" and the devices introduced to solve it, especially external economies,¹ the most debated-over device in the years immediately following Marshall. Then, in the core of the essay, we analyze Knight's work in the 1920s. Following that, we connect Knight's work to the *Economic Journal* controversy and, in particular, to the articles by Piero Sraffa (1926), Lionel Robbins (1928), and Allyn Young (1928). We conclude with some remarks on the theoretical consequences of Knight's contribution.

The "Cournot Dilemma," or the Compatibility between Increasing Returns and Competition, from Marshall to Pigou

The most important and intriguing issue discussed in the controversy on costs in the 1920s was the compatibility—or not—of increasing returns and competition, a Smithian classical issue in the history of economic theory and one that Antoine Augustin Cournot (1838) had contributed to fundamentally. Cournot showed that nothing limits the production of a commodity under conditions of pure competition if a firm's marginal cost is falling. He concluded that, consequently, such firms cannot be in equilibrium and a monopoly would result. In Cournot's estimation, increasing returns and competition were incompatible.

Cournot's conclusion had been well known by marginalist economists like Vilfredo Pareto and Knut Wicksell. However, Marshall was certainly

1. It should be remembered that the concept of external economies used by economists in the period 1890–1930 was different from that used nowadays—externality in the sphere of production. The earlier economists considered both price and nonprice effects. Jacob Viner (1931) termed price effects "technological external economies" and nonprice effects "pecuniary external economies." This second term faded out of the field of pure theory after the 1920s, due to the well-known criticism of the pecuniary external economies as a cause of market failure. This criticism was first made by Knight (1924) and Dennis Robertson (1924) and then fully elaborated in the 1940s.

the economist who devoted the most to “Cournot’s dilemma,” as he called it. He challenged Cournot’s conclusion and proposed a theory of competition that tried to solve the “dilemma” of increasing returns and competition and prove that the two were in fact compatible.

Marshall’s theory in its mature form is presented in books 4 and 5 of the *Principles* (see Marchionatti 1992). In book 4 competition appears as a process that fundamentally relies on the “openness of markets.” Firms are no longer assumed to be price takers, and there are elements of partial and temporary monopoly. Marshall describes the growth of a typical firm that takes substantial advantage of internal economies through production on a large scale; it increases its efficiency by exploiting those economies and thus may become a monopoly. In effect, Marshall argues that a firm may grow rapidly as long as factors such as skill, inventiveness, and entrepreneurial energy exist. However, the firm does not inevitably monopolize the market, because its growth is hindered by the rise of diminishing returns in its life cycle and the difficulties of marketing—problems typical of a rapidly growing firm. For this reason, the monopolization of the market by a firm is, at most, partial and temporary. Moreover, the external economies—“those dependent on the general development of the industry”—may also impede monopolization in the growth process of the firm. These are independent of the size of the firm and influenced by “the concentration of specialized industries in particular localities.” In this spatial context external economies shift the advantages in efficiency gained by the more innovative producers to the other firms. As Marshall himself emphasizes, they are partially irreversible. Hence, in book 4 the dilemma seems to be solved.

The problem of reconciliation comes back in book 5 of the *Principles*. Here the theory of value is featured in relation to the normal cost of production of a commodity relative to a *given* aggregate volume of production in a competitive, static (or stationary) context of partial equilibrium. Here, where Cournot’s conclusion would seem inevitable, Marshall introduces the “representative firm”—representative of the average life cycle of the firm in the industry²—as the crucial device in order to

2. Marshall did not introduce external economies in book 5 of the *Principles* as a device to overcome the dilemma, or at least not principally with such an aim in mind. This is a view that some Marshall scholars hold. George Stigler’s (1941) is considered the most authoritative expression of the view that the external economies are the essential device introduced by Marshall in order to permit reconciliation, but we have to stress that Stigler also wrote that Marshall would probably agree with Robertson (1930), who considered the representative firm as the crucial device.

reconcile increasing returns and competition. The concept of the representative firm avoided, according to Marshall, what he calls “Cournot’s error” of abusing abstract reasoning. In this case this error consists in confusing the tendency of the price of a commodity to fall as a result of the industry’s gradual development, on the one hand, and the individual firm’s tendency to rapidly introduce new economies. Marshall’s discussion appears to be an inextricable mixture of static and dynamic components, which, according to him, had the advantage of furnishing a “reasonably” less abstract theory. Yet it also incorporated what Henry L. Moore (1906) defined as the “bewildering vagueness” of the term *competition*, which prevented the theory of competition from being established precisely. The “vagueness” of that term proved to be unacceptable by the economists of the newer generation.³ Moore’s widely held judgment was the main reason for the increasing importance of the concept of external economies, which was seen as a useful device to resolve the dilemma within a stationary context. This was a point of view that neither attracted nor particularly concerned Marshall.⁴

F. Y. Edgeworth’s review (1905) of Henry Cunynghame’s *A Geometrical Political Economy* (1904) places the reinterpretation of external economies in the center of the solution of Cournot’s dilemma in a static or stationary context. Marginal cost, Edgeworth argued, is a function of a particular firm’s output and of aggregate industrial output. It may be rising with respect to the former and falling with respect to the latter: rising marginal cost curves for individual firms would shift downward with a rise in industrial output, leading to a falling supply curve of the industry.

In turn, Pigou, Marshall’s successor in the Cambridge chair of political economy, adopted Edgeworth’s treatment of external economies—external to the firm but internal to the industry—in his *Wealth and Welfare* (1912) and reasserted it in the first and later editions of *Economics of Welfare* (1920). This approach abandoned the interindustrial characteristic of external economies and their irreversibility. This is what

3. Some Marshall scholars (e.g., Neil Hart [1991]) have been claiming that Marshall’s contemporaries and followers did not fully understand the nature of his methodological struggle and thus they failed to solve the problem of the reconciliation of increasing returns and competitive equilibrium. I think this claim can hardly be accepted. Several authors were conscious of the difference between themselves and Marshall from a methodological point of view and considered their departure from Marshall’s method and methodological devices as necessary for solving the questions left open by him in a more satisfying way.

4. For an analysis of this theoretical debate in England, see Marchionatti 2001.

clearly distinguishes Edgeworth's and Pigou's theory from that of Marshall.

Pigou's theory, however, lacked practical relevance, a point noted by the American economist Allyn Young in his review (1913) of *Wealth and Welfare* (1912). Young emphasized that increasing returns are not of the same theoretical character as diminishing returns and are not subject to an analysis of the partial-equilibrium type. Young asserted that Pigou's concept of external economies could not adequately solve the "Cournot dilemma."

In spite of this criticism, the unchallenged authority of Pigou and Edgeworth, buttressed by Marshall's public silence, left the dilemma substantially unresolved, certainly at least in the English academic environs of a Cambridge characterized by a "complacent and insular attitude" (Whitaker 1989, 172; see also Collard 1990). This situation induced the emergence of a position that clashed logically with the (insufficiently defined) postulates of perfect competition. Hubert Henderson's book (1922) represented a summary of the principles of the Marshallian doctrine typical of the Anglo-Saxon world at that time. His book did not deal with the theoretical questions involved while it suggested that, in the long run, cost may decrease and the price fall as a result of a revolution in the productive method. Only at the beginning of the 1920s did most economists come to consider the state of the existing price theory unsatisfactory and to agree that a work of theoretical clarity and reassessment was needed. This work was undertaken largely by the American economist Frank Knight and later by the Italian economist Piero Sraffa. Knight unified different pieces of the criticism, conducted a work of "refinement," as he called it, and tried to identify the means of overcoming the impasse upon which he was to lay the foundations of modern microeconomics. At that time, only Joseph A. Schumpeter possibly equaled Knight in his understanding of both the dilemma and the way to solve it, but his early works in German (1908 and 1912) had evoked only scant resonance in the English-speaking world.

Knight's Contribution to Microeconomics, 1921–25

In his work on microeconomics during the first half of the 1920s, Knight dealt with a number of issues, among them the perfect competition theory, which he reformulated along Walrasian and Paretian lines; the

form of the supply curve in the long run; and the Pigouvian doctrine of external economies, which he dismissed. He also suggested refinements to the theory of monopoly.

A “Refinement” of the Perfect Competition Theory

Knight’s revised doctoral dissertation, *Risk, Uncertainty, and Profit* (hereafter *RUP*), originally presented in 1916 at Cornell under Allyn Young’s supervision, was published in 1921. The book was immediately accepted by a great part of the international community of economists as a new and outstanding contribution to economic theory, and was very soon considered a classic.⁵ In it, Knight examined two principal issues—the definition of perfect competition and the treatment of risk and uncertainty. Traditionally the latter is identified as Knight’s most important specific contribution to economic theory. However, his definition of perfect competition in the first part of *Risk, Uncertainty, and Profit*—a prerequisite for his treatment of the issues of risk and uncertainty—was tremendously important in the origin of modern microeconomics. In this part of the book, and in articles published in major American economic journals between 1921 and 1925, Knight offered a “complete” reformulation of the perfect competition theory, according to George Stigler (1957) and Paul McNulty (1967). Both in *RUP* and in his article “Cost of Production and Price over Long and Short Periods,” published in the same year, Knight defined the concept of perfect competition in rigorous terms and argued for a sharp separation of the theoretical from the empirical as well as of the “static” from the “dynamic” in economic issues.

Chapter 1 of *RUP* contains the following methodological statements:⁶ (1) a body of “pure theory” in economics (an abstract deductive system) is a “very fundamental,” although small, “section” of economic science; it is a “prerequisite” for economics, as theoretical mechanics is for physics (6); (2) the method of economics is the scientific method,

5. Young had predicted this outcome. At the beginning of January 1921, Young reviewed Knight’s revised doctoral dissertation, ready for publication, and wrote from Harvard: “As I think I have told you before, this is by all odds the ablest thesis that has ever passed through my hands. I suspect that it ought to go a long way towards establishing your reputation as an economist. I have read it with admiration and enjoyment, and really got quite a good deal of intellectual stimulus from it” (letter of 5 January 1921, in Knight Papers, box 62, folder 24).

6. Knight expanded his methodological comments in “The Limitations of Scientific Method in Economics,” a chapter of a 1924 collection of essays about the American discussion on institutionalism edited by Rexford Tugwell.

that is, “the [Paretian] method of successive approximations” (8); and (3) the first approximation of the competitive system—the “particular form of organization of human want-satisfying activity which has become prevalent in Western nations”—is a perfectly competitive system “in which the multitudinous degrees and kinds of divergences are eliminated by abstraction”: it presents the “general principles” of the competitive system (9). Knight maintained that economists had always followed the methodology implicit in those three statements, although not “in a sufficiently self-conscious, critical, and explicit way” (10). Instead, they had employed a “loose use of assumptions and looser application of conclusions” (14). Knight conceded that “notable progress in the right direction” had “recently been made.” This direction was one “toward a sharper separation of the theoretical portion of economics from the empirical portion, and toward the clearer formulation of premises” (14). He attributes this progress to “the work of the mathematical economists and non-mathematical pure theorists,” notably Pareto⁷ and Philip Wicksteed, who had built up “a considerable and fairly satisfactory body of consciously and rigidly ‘theoretical’ . . . doctrine” (14).⁸ Nevertheless he adds: “Unfortunately [the work of pure theorists in general] has not achieved the recognition and been accorded the fundamental place in the general program of the science which we think it should have. . . . In the great mass of economic literature there is certainly still wanting the evidence of a comprehensive grasp of general principles and even more of the meaning and importance of general principles in a scientific program. There is still a need for thoroughgoing and critical comparison and contrast of theoretical assumptions with the conditions of real life and of theoretical conclusions with concrete facts” (14).

7. These methodological statements were influenced by the study of John Stuart Mill’s and Pareto’s works. Pareto’s methodological influence on Knight also appears throughout the book, for example in this note at the beginning of the third part: “The writer is . . . a radical empiricist in logic, which is to say, as far as theoretical reasoning is concerned, an agnostic on all questions beyond the fairly immediate facts of experience” (201 n). On Pareto’s methodology and its meaning for modern economics, see Marchionatti and Gambino 1997 and Marchionatti 1999.

8. Another author with whom Knight was in sympathy with regard to pure theory is Gustav Cassel, whom he considered “one of the leaders of thought in the field of economic theory” (Knight 1921a, 145). In his review of Cassel’s *Theoretische Sozialökonomie*, Knight emphasized that the work “is characterized throughout by keen analysis, rigorous deduction and carefully accurate statement. . . . His procedure is the familiar and unquestionably correct one of beginning with extremely simplified conditions and adding complications by stages, approaching by successive steps to the intricacies of actual phenomena” (145).

On this subject, Knight noted that an “indication of progress in this field is furnished especially by the discussion centering around the concept of normality in the work of Marshall in England and the related notion of the static state espoused in particular in this country by J. B. Clark” (15). However, Knight considered the role of Joseph Schumpeter to be more important. He noted that “the static state idea is further developed along rigidly theoretical lines by Professor Schumpeter in Austria” (15 n. 1). Knight correctly identified the theoretical originality represented by the two books (1908 and 1912) written by the young Austrian economist then available only in German.⁹ Schumpeter expressed the necessity of exposing the canons of pure economics along Walrasian lines. He maintained that pure economics is a field of the application of exact logic and hence adopts the static method to examine the problem of equilibrium, the core of which is the theory of price. In this context he deeply criticized Marshall, holding that the acclaimed Marshallian theory of value was eclectic and “old-fashioned” and it did not achieve the clearness and unity of Walras’s and Pareto’s work in its attempt to keep some aspect of classical conceptions. Schumpeter acknowledged that Marshall’s use of the biological analogy was justified in his attempt to understand the phenomenon of economic growth, but he did not consider this a fruitful path. Instead Schumpeter proposed a fundamental difference between statics and dynamics and claimed that the system of pure economics was essentially incompatible with the problems of growth. He urged the use of the old mechanical analogy describing a state of rest—which induces the use of exact methods and so permits a correct knowledge of things.

Knight adopted Schumpeter’s critical judgment of Marshall but did this on more respectful terms than other critics.¹⁰ He recognized that the

9. Knight refers explicitly to Schumpeter’s *Theorie der Wirtschaftlichen Entwicklung* (1912), but he probably also knew Schumpeter’s *Das Wesen und der Hauptinhalt der Theoretischen Nationalökonomie* (1908). In fact, Knight had an excellent knowledge of German-speaking economists.

10. Regarding this, Knight (1921a, 147) wrote against Cassel: “Professor Cassel seems rather to put himself out to contrast his position with that of the classical economists, and Marshall in particular. In most instances the reviewer is in sympathy with the author’s position. . . . But the real issue is not usually vital if allowance is made for crudity of statement by the earlier writers. Especially it is not necessary to disagree sharply with Marshall in order to agree with Cassel where he is criticized by the latter. As everyone understands, Marshall shows a deliberate preference for cautious, not to say ‘loose’ forms of expression, on the ground that they are more descriptive of reality. The world is like that rather than like the rigorous, sharply definite propositions of the logician. One who disagrees with this conception of the method of theory, as does the present writer . . . should still appreciate the force of the argument in its favor.”

meaning and bearings of the fundamental concepts were much better worked out by Marshall than by any other generally read writer. However, he argued that Marshall had adopted a cautious, almost antitheoretical, attitude toward fundamentals. Marshall had refused to lie down and follow rigidly defined hypotheses. He insisted on sticking as closely as possible to concrete reality. Knight emphasized that Marshall erred when he discussed “representative” conditions rather than “limiting tendencies.” Knight considered the outcome of Marshall’s attitude a theoretical failure: “The gain in concreteness and realism is in our opinion much more than offset by the obscurity, vagueness, and unsystematic character of the discussion, the inevitable consequence of burying fundamentals in an overwhelming mass of qualification and detail” (15).

Consequently Knight thought it necessary to adopt “a sharper separation of the theoretical portion of economics from the empirical portion, and toward the clearer formulation of premises” (14). In this way Knight identified static method and equilibrium as the two key concepts of pure theoretical analysis. First of all, he stressed the necessary scientific role of the static method in economics, the only nonarbitrary assumption (see also Knight 1921b). He maintained:

The opposition to pure theory in general is based on a failure to understand it, and especially common is the misconception as to the meaning of static or normal hypotheses. It is not recognized that their use is inherent in the methodology of science, is in fact the very essence of scientific procedure. . . .

The “static” method in economics does merely this. It inquires what conditions exist and studies the results which recognizable forces at work . . . tend to produce under those conditions. (16)

The other key concept, that of equilibrium, Knight said, is closely related to the concept of static method and, likewise, is methodologically necessary:

It is the nature of every change in the universe known to science to have “final” results under any given conditions, and the description of the change is incomplete if it stops short of the statement of these ultimate tendencies. Every movement in the world is and can be clearly seen to be a progress toward an equilibrium. . . . Every change is an equalization of the forces which produce that change, and tends to

bring about a condition in which the change will no longer take place. . . .

So also in economic phenomena. (17)

Once Knight set down these assumptions, he moved on to the study of a perfectly competitive society in part 2 of the book (chapters 3–6). He began by noting (in chapter 3) that “the historic body of economic theory rests upon the assumption of perfect competition.” However, “the precise character of this assumption has been partially implicit and never adequately formulated.” Hence “the assumptions actually made and their implications need to be brought to the surface and emphasized” (51). Knight considered this the basis needed to advance and explain the problem of profit. Knight continued: “In order to study first the most essential features of exchange relations, it will be necessary to simplify the situation as far as possible by a process of ‘heroic’ abstraction” (76). Perfect competition is conditioned by the existence of a set of assumptions, the most important of which are the following: (1) “a perfect market for productive services . . . , that is, uniform prices over the whole field” (1921b, 316); (2) complete rationality and perfect knowledge by free and independent individuals; (3) “perfect mobility in all economic adjustments, no cost involved in movements or changes” (1921c, 77); (4) “virtually instantaneous and costless” exchange of commodities (1921c, 78); (5) “perfect, continuous, costless intercommunication between all individual members of the society” (1921c, 78); (6) perfect divisibility of commodities; and (7) “an indefinitely large number of competing organizations, *each of the most efficient size*” (1921b, 316). It is important to note that these very restrictive assumptions are closer to our contemporary definition of perfect competition than any other up to that time: Knight’s definition implied a conceptual change with regard to the classical notion of competition, one totally devoid of behavioral content (see McNulty 1967).¹¹ On this basis, Knight was able to approach the Cournot dilemma.

11. It is fair to remember that, in the United States, Moore (1906) had already approached Knight’s rigor in defining the premises of free competition. Moore refers to Pareto, the economist who had most of all contributed to the foundation of this line of thought from the methodological and analytical points of view. Moore identified the following premises: (1) every economic factor seeks and obtains a maximum net income; (2) there is only one price for commodities of the same quality in the same market; (3) the influence of the product of any one producer upon the price per unit of the total product is negligible; (4) the output of any one producer is negligible compared to the total output; and (5) each producer regulates the amount of his output without regard to the effect of his act upon the conduct of his competitors.

Incompatibility Reaffirmed

In chapter 4 of *RUP* Knight dealt with the *vexata questio* of the form of the supply curve in the long run. He recognized that “the question is really one of the most difficult and perhaps one of the worst muddled in economic theory” (121). He maintained, however, that “examination *seems* to show that *under the conditions necessary to perfect competition, costs must always increase as supply increases*” (121; italics added). The *necessarily* increasing shape of the supply curve under conditions of perfect competition is explained in *RUP* and, more widely, in his 1921 article as follows:

Under the conditions assumed [free competition], an increase in the production of any commodity means a transfer of productive resources into the industry and a decrease in the production of some other commodity. But, other things being equal, this decrease in the production of other goods will raise their prices and increase the strength of the competing attraction which they exert on productive resources against the industry in question in which output is being increased. *In simpler terms, an increase in the output of any industry involves increased demand for the productive goods used in it, which increased demand raises their prices, that is, raises the costs of production of the commodity turned out.* (315–16; italics added)

“It goes without saying,” Knight wrote in *RUP*, “that the cost of all units of the supply is the same” (122). In fact, as his footnote adds, “the existence of differences in costs in different establishments in an industry is proof . . . that the competitive adjustment is imperfect” (122 n. 1).

Perfect competitive adjustment of an industry, he argued, is characterized by technological and demand conditions that enable a large number of organizations to remain in that industry. Under competitive conditions,

all establishments will be forced to the most efficient size, and variation in the output in the industry means a change in the number of establishments, without change in their scale of operations individually. . . .

The specification of a plurality of establishments each of the most efficient size eliminates at once both the possibility of decreasing costs

due to increased efficiency under larger output and also the entire notion of marginal costs. (1921b, 316–17)

“If increased economies are available through larger-scale operations,” Knight concluded, “then larger-scale operations will be introduced under competition, through an increase in the size of the establishment with a *reduction in the number* of establishments and *without an increase in the output of the industry as a whole*. The tendency to an increase in size and reduction in the number of establishments will go on . . . until either all establishments reach a size of greatest and equal efficiency (not necessary equal size) or else until there is only one establishment left in the industry” (317).¹²

Here Knight was able to criticize the “defective analysis” that was widespread at that time. In a 1923 review of “some books on fundamentals” he criticized Henderson’s *Supply and Demand* (1922), which represented Cambridge economics, “the modern British ‘neo-classical’ interpretation of economic science founded and mainly developed by Dr. Alfred Marshall” (342); here, Knight once again confronted the problem of “increasing returns and competition.” He maintained that Henderson’s discussion of the meaning of cost of production revealed “uncertainty and confusion” (345) and summarized his argument as follows: “The author holds that, for short periods, there is a ruling tendency toward increasing costs as supply is increased, and hence toward an increase in price with an increase in demand, but that, in the long run, costs may decrease and the price fall, as a result of a revolution in productive methods” (345). Knight remarked that “this is perhaps the position most commonly taken, but it can be shown to rest upon defective analysis” (345) and argued,

There is no direct casual relation between increased output and *new discoveries* in the line of productive methods, and hence none between increased output and reduced costs from this source. As to economies of large-scale production by methods already known, the tendency

12. With regard to large-scale operations, Knight noted “an error very different from that of treating price-determining cost as a decreasing function of output but not unconnected with it and very common” (334), namely “the exaggeration of the economy of large-scale production” (334). It is true that “a spectacular saving is effected in certain operations,” he wrote, but “to make that saving possible large organizations must exist and the cost of internal cohesion in large groups of men is very high” (334). This observation is similar to that made by Pareto (1896–97).

should be to secure all of these that are available, by increasing the size of establishment in the industry, independent of any increase in the aggregate output, through reduction in the number of establishments. If competition is effective this tendency should operate . . . until all the establishments are of the most efficient size. . . .

Under ordinary competitive conditions, the long-run tendency must run toward increasing costs since increased production means an increased demand for the cost goods and higher prices for these. (345–46)

Therefore, Knight concluded, the argument for a long-run tendency toward decreasing costs with increasing aggregate output “either assumes conditions incompatible with competition in the industry or else involves a confusion between ‘static’ and ‘dynamic’ considerations” (345).

Young had already criticized Knight’s argument in his 1921 “notes”:

You miss the point, I fear, of Marshall’s external economies. These are economies (in general) of greater specialization or division of labor. They come with an increase of output in the industry, and need not presuppose (a) any change in industrial knowledge or technique; (b) any increasing size of plants. An increased output means new plants, of course, but the important thing is that they are not “similar establishments,” but, in general, more highly specialized establishments. Whether the price goes up . . . is a matter of whether one sort of influence or another . . . prevails. It is an empirical matter,—not to be reached from general principles. (Knight Papers, box 54, folder 14)

In the analysis of the possible coexistence of increasing returns and competition, Knight had not yet considered the concept of external economies, probably because he had accepted Young’s criticism of Pigou in 1913. At Harvard, however, where Young went in 1920, Young had begun to become more and more disaffected and skeptical toward the model of static equilibrium. He had begun to develop the idea of external economies as something that would provide a useful device to resolve the dilemma along the lines that he was to develop in his 1928 article (see Blicht 1995). To Young’s repeated criticism (see folder 16 of the Knight Papers), Knight provided an answer in his 1924 article in the *Quarterly Journal of Economics*, where he criticized Pigouvian external economies.

The Criticism of Pigouvian External
Economies, 1924–25

In his famous 1924 article “Some Fallacies in the Interpretation of Social Cost” and in a successive 1925 article, Knight dismissed the Pigouvian idea of external economies as used by Pigou (1920) and Frank Graham (1923). Knight reasoned that, under conditions of increasing returns, an industry tends to become monopolized:

Decreasing cost (or increasing returns) is alleged to result in several ways, which can be dealt with but briefly. The most important is the technological economy of large-scale production. When the output of a commodity is increased, the cost of the productive services used to produce it will be higher; but this increase in their cost per unit may, it is held, be more than offset by economies in utilization, made possible by larger-scale operations, which increase the amount of product obtained from given quantities of materials and resources consumed. . . . *If competition is effective, the size of the productive unit will tend to grow until either no further economies are obtainable, or there is only one establishment left and the industry is a monopoly. When all establishments have been brought to the most efficient size, variation in total output is a matter of changing their number, in which no technical economies are involved. . . .*

These brief statements form a mere summary of the argument that, with reference to long-run tendencies under given general conditions, increasing the output of a commodity must increase its cost of production unless the industry is, or becomes, a monopoly. (1924, 596–98; italics added)

Furthermore, Knight maintained that “the doctrine of ‘external economies’ . . . rests upon a misconception”: “Economies may be ‘external’ to a particular establishment or technical production unit, but they are not external to the industry if they affect its efficiency. . . . *External economies in one business unit are internal economies in some other, within the industry*” (597; italics added). It thus followed that “any branch or stage in the creation of a product which offers continuously a chance for technical economies with increase in the scale of operations must eventuate either in monopoly or in leaving the tendency behind and establishing the normal relation of increasing cost with increasing size” (597).

Knight's objection to the concept of external economies was based on his observation that what he called "pure external economies" simply do not exist. By "pure external economies" he meant the type of external economies compatible with conditions of perfect competition, in particular with the static equilibrium hypothesis. Young (1913) had already suggested this, as also did Dennis Robertson (1924).¹³ In his rejoinder to Frank Graham's reply to "Some Fallacies," Knight (1925) wrote that he was skeptical about the existence of the kind of "purely external economies in organization" required by the Pigou-Graham argument: "In spite of the weight of authority which may be cited for such economies, I have never succeeded in picturing them in my mind, or finding any convincing reason to believe they exist. I can imagine conditions in which the production of a good may be tied up with a monopolistic industry like transportation, operating below capacity, in such a way as to produce some tendency in the direction argued. *But I cannot believe such conditions general enough to justify a special law in economic theory*" (332; italics added). He concluded forthrightly that "until a plausible example is brought forward, the category of decreasing cost under stable competition remains an 'empty economic box'" (333).

The logical criticism of Pigouvian external economies was reconsidered in 1926 in a long review of the second edition of Pigou's *The Economics of Welfare* (1924), which Knight then regarded as the science of "economics at its best." Knight insisted that "the notion of external economies does not stand up well under examination" (56) and reasserted his 1924 thesis:

Much discussion of industries of increasing and decreasing returns is vitiated by an incorrect assumption as to the definition of an industry. For the purpose of an argument about apportionment of productive power among industries *the term must be understood to designate the entire process of producing a finished product. The same basis is required for distinguishing between external and internal economies.* An economy accruing to another stage of the production process in consequence of expansion in one stage is not external but internal. And any internal economies involve a tendency toward monopoly of whatever

13. The statement "external economies in one business unit are internal economies in some other, within the industry" has seemed doubtful to some authoritative authors (for example John Chipman [1965] and Stigler [1987]). However, I think that the statement is unquestionable from the point of view of Knight's rigorous theoretical premises *and* practical relevance, that is, his methodological assumptions.

part of the industry is subject to them, and toward the segregation of the saving as monopoly return. (56; italics added)

Increasing Returns and the Theory of Monopoly: Some Suggestions

Knight's analysis was of major consequence, as it reaffirmed Cournot's conclusion that increasing returns and competition were incompatible. Knight (1921b, 332) emphasized that "we have only argued that such a cost curve is incompatible with long-run competitive conditions" and identified the boundaries within which this conclusion was valid: "All of this reasoning relates to the ultimate goal of the competitive tendencies, with *unlimited time* allowed for the adjustment of production to given conditions of demand" (320; italics added). At the same time he stressed that markets in the real world are never purely competitive or monopolistic, but a mixture of the two, and that decreasing costs was the normal condition for firms operating in that state: "One of the most serious oversights in the discussion of decreasing cost is the neglect of the mixture of competition and monopoly which is a general characteristic of the type of business supposed to exhibit this type of cost function" (332).¹⁴

This opinion—that industry was to a large extent in an intermediate state between monopoly and competition—was of course not new in the United States. Henry L. Moore (1906) strongly expressed a similar view, later shared by economists like Jacob Viner (1921) and J. M. Clark (1923). However, Knight undoubtedly went further and offered some positive insights similar to those developed some years later by Piero Sraffa and the theoreticians of imperfect and monopolistic competition. In Knight's theoretical framework the competitive theory represents a first approximation. In order to make economic theory more realistic, it may be supplemented with a second approximation, represented by a partial monopoly theory.

In his 1921 article Knight (1921b, 332) made some interesting suggestions on the analysis of this second approximation. "In the case of partial monopoly," he writes, "the correct approach to the explanation

14. Knight wrote: "Just as part of the traffic of a railroad is competitive and part monopolistic, nearly every manufacturing and mercantile business has a monopoly on *some feature* of its product; its good or service is differentiated from others in some manner and to some degree. To the extent that any business is monopolistic it may manifest decreasing costs due to the 'economy of large-scale production'" (332).

of price . . . would seem to be to apply the theory of monopoly, not that of competition.” Knight suggests, “Instead of attempting to allow for a degree of monopoly in the supply . . . it is vastly simpler to allow for partial competition as a phenomenon of *substitution*, on the demand side. No difficulty whatever is involved in assuming control of the supply (of the commodity defined in the narrowest sense) and allowing for competition by substitution of more or less similar goods in drawing the demand curve. And this is the more realistic view as it represents the way in which the producer would naturally envisage the situation” (332).

Knight did not contribute to the reorientation and refinement of the analysis of monopoly that involved many economists after 1925. When he eventually did take up this subject many years later, he criticized the evolution of the theory in the field (see Patinkin 1973, 794). In a 1939 paper he sustained a position, then widely held at the University of Chicago, that the theory of imperfect competition could not be considered revolutionary. He argued that it was largely just a theory of monopoly and that Cournot had already said most of what was important on the subject. Nevertheless, Knight recognized that some analytical progress had been achieved, particularly in the use of the curve of marginal revenue in relation to marginal cost and both in relation to average curves.

Knight and the Cost Controversy in England

After nearly a decade of theoretical silence, the controversy surrounding Marshall’s theory of competition started up again in England in 1922–24. It centered on the debate on “empty economic boxes” among John Clapham, Pigou, and Robertson. The controversy reached its peak on the pages of the English *Economic Journal* in the second half of the 1920s. Its mature phase was marked by a continuing dialogue of articles. Sraffa (1926) launched an explicit and radical criticism of the Marshallian theory of competition.¹⁵ Pigou (1927, 1928a)¹⁶ and Robertson (1930) answered Sraffa with Marshallian arguments. Robbins (1928) strongly contributed to the shattering of Marshallism in England. Young (1928), Schumpeter (1928), and Shove (1930) showed their disaffection with the model of static equilibrium and stressed the dynamic aspects

15. For an analysis of Sraffa’s articles of 1925 and 1926—the first part of the 1926 article was a partial summary of the 1925 article—see Marchionatti 2001.

16. Taking off from Pigou (1928a), Viner (1931) offered a representation of the reconciliation between increasing returns and competition that has been considered a classic.

of Marshall's theory. Three of these articles are the most relevant to the present essay—those of Sraffa, Robbins, and Young—the first two because they reveal strong and striking analogies with Knight's writings, Young's because it was a subject of a private debate between Young and Knight.

The core of Sraffa's famous 1926 article examined to what extent Marshall's supply curves could be used in the study of the equilibrium value of single commodities under competitive conditions. In book 5 of the *Principles* Marshall assumed that the conditions of production and the demand for a commodity can be considered as being practically independent with respect to small variations, both in regard to each other and in relation to the supply and demand of all other commodities. Sraffa considered this assumption of Marshall illegitimate. He allowed for it to be legitimate if a variation in the quantity produced by the industry under consideration has no, or only slight, collateral effects. However, the applications of the laws of returns in the great majority of cases fall into the case where the collateral effects are such that they upset the conditions of the particular equilibrium that was meant to isolate those effects. In regard to the decreasing supply curve, Sraffa believed that there was a way of escaping the Cournotian conclusion regarding the problem of "increasing returns and competition": to use the concept of the external economies à la Pigou, but to keep in mind that these economies constitute the class most seldom encountered. Sraffa's 1926 article had a dramatic impact on the Cambridge academy. His logic appeared conclusively "destructive" (Keynes 1930). His outline of a theory of imperfect markets greatly impressed the old and new generation of English economists.

Robbins's article also had a relevant impact on the English debate. Robbins attacked the representative firm and held it unnecessary for the study of stationary equilibrium. According to Robbins the representative firm was a poor tool for examining the problems of change. He also questioned the existence of external economies.

Sraffa's and Robbins's critiques emphasized the deficiencies of the Marshallian approach from the analytical point of view. They pointed out that there was inconsistency between logic and practical relevance. They both followed a line of rigorous conceptualization, as did Knight. Sraffa's criticism, in particular, presents strong analytical similarities with the American economist. Both treat perfect competition as a first approximation; both take the rigorous notion of equilibrium from Pareto and Enrico Barone; both criticize the Marshallian solution to the Cournot

dilemma of increasing returns and competition; both offer positive insights on imperfect competition. The analysis of each seems to be variation on a common theme. Lastly, Robbins—and seemingly Sraffa too—adhered to the necessity of a separation between statics and dynamics, much like Schumpeter and Knight.¹⁷

These analogies make it natural to ask the question: Did Knight's work in the years 1921–25 influence the English debate? We have scarce evidence of such an influence. Sraffa himself said that his article was an attempt to coordinate previous critical material. However, we have no evidence at all from his writings and unpublished notes that he knew of Knight's papers while he was writing his articles. Here we may be facing a phenomenon not unusual in the history of ideas—the appearance of similar theoretical positions in different places independently from each other. Robbins's position is slightly different. In his article, Robbins questions the existence of external economies and, in so doing, refers to Young's (1913) and Robertson's (1924) criticisms, which were strictly connected to Knight's criticism in his 1924 article. It is not improbable that Robbins, a young economist at the London School of Economics, knew Knight's article. In his autobiography (Robbins 1971) he referred to his meetings with Young when Young was at the LSE in the years 1927–29; Robbins tells of his personal relations with “most of the more interesting economists of the day” (132)—Knight and Viner from Chicago among them—in the years following. We may suppose that Robbins had known Knight's writings but preferred not to quote a critic of Cambridge economics in his *Economic Journal* article. Instead, he quoted Young's and Robertson's papers because they were widely known and appreciated in the English debate.

Another natural question arises: Was Knight involved in the English debate after 1926? In 1928 Knight published an article on the general theory of price in the *Journal of Political Economy*. In it he referred to Sraffa's article on the importance of cost as the “cause” of price. He

17. On the basis of a letter exchange between Schumpeter and Sraffa in spring 1930, we may argue that Sraffa must have also shared the opinion that the separation between statics and dynamics was necessary. Sraffa first sent Schumpeter a reprint of the *Economic Journal* symposium. Schumpeter replied in a letter dated 3 May 1930 (Sraffa Papers, folder C281.1). He was convinced that Marshall's weakest point—the analytical difficulties connected to the representative firm and the laws of costs—centered on Marshall's inability to distinguish between the phenomena of static equilibrium and those connected with the industrial process. In his reply Sraffa described Schumpeter's considerations as an “illuminating remark” (Schumpeter Papers).

devoted a section of it to the issue of the shape of the cost curve, where he reasserted his criticism of Pigouvian external economies and shortly debated with “the Cambridge economists.” This article, as a contribution to the cost controversy, has escaped the historians of economic thought as well as—strangely enough—his contemporaries. Knight considered the cost controversy also in private correspondence with Young. Actually, he was directly involved in the discussion of Young’s 1928 paper, “Increasing Returns and Economic Progress.”

In his article Young considered external economies as the prime source of increasing returns and economic progress, interpreted as a part of a theory of industrial growth within the perspective of book 4 of Marshall’s *Principles*. The central point of Young’s article was that Marshall’s partial analysis of book 5 obscured certain aspects of the process of industrial progress. Young maintained that “the internal economies of some firms do not account for all the external economies.” This is a clear reference to Knight’s 1924 criticism. Thus the traditional apparatus—which assumes a condition of comparative stability in looking at the internal economies of a particular firm and neglects changes like the appearance of new products and new industries and the firms’ assumption of new tasks—had to be discarded. The issue had been the object of an ongoing discussion between Young and Knight for a long time (see above, “Incompatibility Reaffirmed”). This is reflected in letters and manuscript notes as well as, indirectly, in Knight’s 1928 article. Knight received the proofs of Young’s 1928 presidential address (which was presented three months before its publication in the *Economic Journal*) and sent Young his 1928 article. Then, in a letter to Young of 24 September 1928 (Knight Papers, box 62, folder 24), Knight wrote that in his paper he had tried to indicate “what I increasingly feel to be the main issue between my position and the ‘Cambridge’ one, namely the making of a sharp separation between the ‘static’ and ‘dynamic’ problems.” In his 1928 paper Knight had in fact written, “The Cambridge economists are averse to making a clear separation between adjustments to given conditions and effects of change in given condition (resources and property rights, wants, and methods)” (367 n).

In the letter Knight emphasized that his “negative feelings towards diminishing costs” relate “exclusively” to a static situation. Knight believed that Marshall, Pigou, and Young himself had not made the rigorous separation that, on the contrary, he considered necessary. Without this separation, as he wrote, “I do not see how we can know what we

are talking about and have a clean-cut theory in any other way.” He recognized that it was a purely theoretical matter and that empirically “no such line can be drawn.” However, he underlined the fact that “to the purposes of theory” it seemed to him “essential” “to get completely rid of the time variable” in order to avoid the theoretical confusion that the Cambridge economists fall into.

Knight thought that a rigorous separation between statics and dynamics was vital for overcoming the imprecise concept of competition used by Marshall and for avoiding theoretical misunderstanding. From this perspective, the Marshallian concept of external economies had to be regarded as something that falls within the dynamic area and considered logically inconsistent with the static price theory. Evidently, Young did not fully recognize Knight’s point, but I think that the difference between the two authors was one of accentuation, and not anything really conceptual. In fact, in his letter Knight spoke of “apparent disagreement.”¹⁸ In Young’s reply of 6 October 1928, he does not reject Knight’s argument but writes that “the purely static view does not interest me very much. . . . We have to depart from it somehow. The only question is just how” (Knight Papers, box 62, folder 24). It seems clear that Young also considered external economies to be a subject of a dynamic framework. However, Young was noticeably more sympathetic toward Marshall than were Knight and Schumpeter. Schumpeter’s 1928 *Economic Journal* article again stressed the need for a clear-cut separation between statics and dynamics.

In his letter to Young, Knight made an interesting and important point about the term *dynamic*. He writes that when a sharp separation between statics and dynamics is proposed, the term *dynamics* represents “a serious misnomer.” Instead, he would prefer to use “some such term as ‘evolutionary’ and employ the word dynamics as in mechanics to refer to forces and velocities of change towards an equilibrium under given conditions.” Actually, the discussion of this subject may be considered Knight’s final contribution to the cost controversy. In a German article in the 1930 *Zeitschrift für Nationalökonomie* Knight asserted that the typical use of the terms *statics* and *dynamics*, which he dated back from

18. Knight writes: “Your discussion puzzles me especially because you admit the accumulation of capital into an otherwise static situation. Not that I object in the least to this, or to any of your argument; but I cannot see that it affords any basis for disagreeing with my own position! I wonder if any of the above makes headway towards reconciling the apparent disagreement?” (letter of 24 September 1928, Knight Papers, box 62, folder 24).

Marshall, is confusing. He wrote: "In actual usage economic dynamics, or dynamic economics, has become merely a critical and negative term to refer to the limitations of 'static' analysis, or more exactly to any particular author's objections to any other author's use of equilibrium concepts. Its least vague usage is that of a sort of catch-all for stressing changes in given conditions in contrast with adjustments to given conditions. . . . What it calls dynamics should be called evolutionary or historical economics" ([1930] 1999, 154).

Knight starts off with a discussion of the concept of equilibrium in economic statics and the involved notion of force. He goes on to consider the process by which such an equilibrium is established—that is, the field of dynamics. Here the fundamental question is "whether the problem of economic evolution or progress can be given the form of a problem in dynamics" (155). Knight concluded that in the field of economic progress the notion of tendency toward equilibrium is definitely inapplicable, and he adjudged that in discussing social evolution "the use of the mechanical analogy, the categories of force, resistance, and movement," have to be rejected (168).¹⁹

19. This statement seems to indicate that in the years 1928–30 Knight abandoned the method of successive approximation that characterized his *RUP*. Instead, he adopted a method that suggests that equilibrium theorizing is strictly static and that economic progress requires a historical, rather than a mechanical, investigation. I am indebted to a referee who stressed that Knight argued in his 1930 article that no process of relaxing assumptions, as happens in the method of successive approximation, permits us to move from equilibrium theorizing to reality. See Emmett 1999.

In the year around 1930, just after Knight's return to the University of Chicago in 1928, Knight was rethinking and working on a new refinement of his approach to the theory of cost. Essentially, this represents a qualification and correction of the Austrians' and Wicksteed's cost theory. To a great extent, it seemingly was brought on by Viner's 1932 article on cost theory in the *Weltwirtschaftsliches Archiv*. As Knight himself wrote in 1934, this article "led to some discussion of that subject in the theory group at Chicago" (Knight 1934) and made it necessary for him "to straighten out" his ideas on the subject of cost. Knight and Viner—at that time the two leading members of the Chicago Department of Economics—had different opinions on various points of theory, particularly over the relative correctness of the "real costs" and "opportunity costs" approaches to the theory of value and over the importance of Ricardo. An early example of their different opinion on the theory of cost that anticipates the above mentioned article is a short unpublished critical note on Viner's article that Knight wrote in 1932 (Knight Papers, box 62, folder 14). Knight's anti-Viner interpretation of Ricardo is fully presented in Knight 1935b. The result of Knight's rethinking was "a considerable reconstruction" of his own theoretical system. He presented this briefly in his review of the reprint of Wicksteed's *Common Sense* in 1934. He worked this out more fully in a German article published in two issues of the *Zeitschrift für Nationalökonomie* in 1935. These articles constitute the final exposition of Knight's economics since the 1930s, but are outside the scope of this essay.

Concluding Remarks

The construction of post-Marshallian modern microeconomics as a conceptually new field, distinct from the classical and Marshallian one, has remained incompletely assessed in the history of twentieth-century economic theory. A widely shared interpretation holds that the “controversy on costs” on the pages of the *Economic Journal* is crucial for the process of this construction and is a premise for subsequent theories of imperfect competition, whose emergence at the beginning of the 1930s is considered to be a kind of watershed between the old and the new. Here I have tried to offer a different interpretation of that key moment, one in which the controversy on costs appears only as an episode, albeit an important one, in the process. Instead, I have particularly emphasized the methodological contribution of Frank Knight, especially between 1921 and 1925. The logical starting point of Knight’s argumentation was the criticism of the then current method of most economists—essentially, the Marshallian school, dominant in the English-speaking world. According to Knight, the Marshallian method was based upon the “loose use of assumptions and looser application of conclusions” as well as the unclear separation between the theoretical and empirical portions of economics. Marshall’s definition of competition appeared theoretically vague and confused. Knight believed that a different methodological approach, that of the Paretian sort, had to be followed in order to get beyond those limits. In such a framework, the perfectly competitive system—rigorously defined using the two key concepts of static state and equilibrium—constitutes the central subject of pure theory in economics. It is a small but fundamental section of economic science that should be considered a prerequisite of economic theory.

Several consequences of Knight’s approach are noteworthy: (1) the distinct separation between statics and dynamics as two different fields; (2) the reaffirmation of the correctness, within the limits of statical analysis, of the incompatibility of increasing returns and pure competition; and (3) the consequent rejection of the device of external economies used by the Marshallian economists to solve the Cournot-Marshall dilemma, because it was problematic logically as well as unrealistic empirically. Increasing returns and external economies must be considered an essentially dynamic or, better, evolutionary phenomenon, to be studied in a different field. In this way, Knight and the economists following the same methodological path could overcome some of Marshall’s analytical

difficulties and reject the solutions offered by Marshall's followers by maintaining that they were confused, or wavering, or incapable of satisfying the twin requirements of logical rigor *and* practical relevance.

Schumpeter, Sraffa, and Robbins were the other economists who attacked Marshallism in the name of rigor in the 1920s (Schumpeter had already begun in the previous decade). Their criticism presents strong analogies to Knight's. Schumpeter considered Walras's theoretical framework the most advanced: Schumpeter called Marshall's theory of prices old-fashioned and eclectic. He maintained that the system of pure economics was essentially incompatible with the problems of growth. Hence he emphasized the need to separate statics from dynamics in order to overcome Marshall's impasse. Knight was greatly indebted to Schumpeter's criticism of Marshall.

The success of this new approach can be understood by placing it within a historical context of increasing intolerance for the vagueness of economic assumptions. The process of analytical "refinement" and conceptual change instigated by these economists—the champions of theoretical rigor—must have seemed like light in a sea of darkness to the new generation. In fact this process freed economic theory from the confusion it had seemingly fallen into. According to its authors, an approach based on such a conceptual change would provide the *required* treatment for fields such as growth, entrepreneurial profit, and the nature of the firm, as well as for market structures that were different from pure competition and pure monopoly. As is well known, Knight inquired into the issue of profit in the second part of his 1921 book. This involved the abandonment of the restrictive assumptions of the static state and perfect knowledge. As Ronald Coase (1937) recognized, Knight offered valuable insights into the problem of the existence of the firm. Harold Demsetz (1988) actually identified Knight's book and Coase's article as the two works that changed economists' perspectives on the theory of the firm, releasing them from the "chains of the Marshallian representative firm" and directing them toward the idea of the firm as a problem-solving institution.

The emergence of modern microeconomics in the 1920s was a revolutionary event in the history of economics. Frank Knight's writings were a crucial component in all this. They were a mix of radical criticism and positive insights. Knight's work contributed in an essential way to the conceptual change that signified the passage toward the post-Marshallian era in microeconomics.

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